Kyle Cyree, MBA, MFA

US CITIZEN • NATIVE ENGLISH • RALEIGH, NC (REMOTE)

Principal Product Architect • AI Product Lead & Systems Builder

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EXECUTIVE SUMMARY

Technical Product Leader, Principal Product Architect, and Systems Builder (UNC MBA, Emerson MFA) with 15+ years of experience owning end-to-end product outcomes across AI-native platforms, regulated fintech, enterprise search, and high-scale healthcare. Operates as a high-agency product leader who prototypes directly in code and AI harnesses (Cursor, Claude Code, TypeScript, Next.js, Python, SQL/DuckDB-WASM, Rust) before committing engineering sprints, inspects Git PRs daily, and bridges relational data architecture with commercial execution. Proven track record launching a chartered digital-asset bank from zero to one (Custodia Bank core banking and Bitcoin custody in 6 weeks), rescuing and scaling Fortune 5 healthcare apps (UnitedHealth Group & Optum 35M+ patients, $100M+ Apple Health revenue impact), re-architecting Fortune 1 enterprise search (Walmart Global Tech for 3M+ associates), and driving 10x operational scale and fraud reduction at Foodler ($500M+ volume, 83% chargeback reduction, acquired by Grubhub).

CORE COMPETENCIES & TECHNICAL STACK

Product Leadership & Execution: Zero-to-One Product Discovery & Roadmapping, Autonomous Squad Leadership, AI-First Rapid Prototyping, Relational Data Modeling & SQL Analytics, PR Inspection & Git Workflows, Regulated Fintech & Healthcare Compliance, B2B & Consumer Unit Economics.

AI & Systems Architecture: Multi-Agent Orchestration, Model Context Protocol (MCP), Grammar-Constrained Local LLM Harnesses (Qwen/Llama), In-Browser Analytics (DuckDB-WASM, OPFS, Apache Arrow), Browser-Local Inference (WebGPU, Transformers.js), GraphQL & Content-Addressed Knowledge Graphs.

Technical Stack: TypeScript, Next.js (App Router), React 18/19, Node.js, Python, SQL, DuckDB-WASM, PostgreSQL, Drizzle ORM, Turso LibSQL / SQLite, GraphQL, Tauri 2.0, Rust, Swift / SwiftUI, Kotlin / Jetpack Compose, Tailwind CSS, Cloudflare Workers.

Tools & Methodology: Git / GitHub, Cursor, Claude Code, Gemini API, Figma, Linear / Jira, SQL Dashboards & Experiment Design, Executive One-Pagers & Service Design.

THE "UNIT OF ONE" OPERATING LOOP

01. Executive & Regulatory Strategy: Applies UNC Kenan-Flagler MBA rigor to untangle OCC/FDIC banking compliance, HIPAA clinical workflows, and B2B unit economics before writing a spec.

02. First-Principles Data Modeling: Designs relational schemas (PostgreSQL, Drizzle, GraphQL), local-first DuckDB-WASM engines, and zero-PII cryptographic boundaries before touching pixels.

03. Swiss Grid & Interaction Craft: Builds mathematical Massimo Vignelli layouts and Josef Albers perceptual color systems that make dense institutional telemetry feel effortless.

04. AI-Accelerated Shipping: Uses advanced prompt and grammar-constrained harness engineering to turn architectural specs directly into working Next.js/TypeScript, Python, and Rust software with zero handoff loss.

SELECTED IMPACT HIGHLIGHTS

- **Sovereign Knowledge Protocol & Multi-Platform AI Ecosystem (Pipes Protocol — pipes.pink):** Solo architected and shipped a 4-tier knowledge protocol and marketplace featuring SHA-256 content-addressed context graphs, MCP Agent Gates, the ! BANG macOS Native Desktop IDE (Tauri 2.0 + Rust with local Qwen/Llama grammar-constrained JSON harnesses), native iOS (SwiftUI) and Android (Jetpack Compose) apps, and the My First Bitcoin 31-node curriculum across 90+ languages.
- **In-Browser Zero-PII Civic & Financial Data Engine (Marigold Insights):** Architected a browser-local analytics workbench powered by DuckDB-WASM, OPFS, and multi-threaded Web Workers capable of querying 2,000,000+ records in-browser at 60fps with zero cloud egress and deterministic Fellegi-Sunter probabilistic record linkage (1,000 pairs in 10.2ms).
- **Zero-to-One Chartered Digital Asset Bank (Custodia Bank):** Founding product design and systems architecture lead for a chartered Wyoming SPDI full-reserve bank ($70M+ raised). Shipped a 30-day Federal Reserve core banking prototype and 6-week institutional platform across Fedwire/ACH, multi-sig Bitcoin UTXO custody, and Avit settlement—requiring 0 additional design hires for 18 months.
- **Fortune 5 Healthcare Crisis Rescue & Ecosystem Scale (UnitedHealth Group & Optum):** Stepped in following a team transition to stabilize flagship mobile applications serving 35M+ patients and 400K+ employees, and championed native Apple Health integration driving an estimated $100M+ in revenue impact.

PROFESSIONAL EXPERIENCE

Colon Hyphen Bracket R&D — Pipes Protocol & Marigold Insights | **Principal Product Architect & AI R&D Lead** | **DEC 2018 - PRESENT**

- **Sovereign Knowledge & Agent Infrastructure (Pipes Protocol — pipes.pink):** Architected and built a complete multi-platform knowledge protocol and creator marketplace featuring SHA-256 content-addressed context graphs, MCP / HTTP 402 agent gates, 25 canonical FOSS knowledge pipes, the ! BANG Tauri 2.0 / Rust macOS IDE for local LLM synthesis, native iOS (SwiftUI) and Android (Jetpack Compose) mobile apps, and 90+ language interactive classrooms for My First Bitcoin.
- **Zero-Egress Analytics & Dual-LLM Systems (Marigold Insights, Superstar, Greater-Than):** Built Marigold Insights (in-browser DuckDB-WASM + OPFS engine processing 2,000,000+ records with Fellegi-Sunter probabilistic linkage and zero PII cloud leakage), Superstar (Dual-LLM Gemini + DeepSeek reasoning bridge), and greater-than (open-source WebGPU + Transformers.js support agent with $0 server inference overhead).

Custodia Bank, Inc | **Lead Product Designer (First Design Hire)** | **JAN 2022 - SEP 2024**

- Founded the product design and UX systems practice as first design hire for a chartered Wyoming SPDI full-reserve digital asset bank ($70M+ raised), scaling from initial wireframes to a full institutional banking and custody suite.
- Delivered a complete company rebrand in 4 days, a core banking application prototype for Federal Reserve regulators in 30 days, and a comprehensive institutional design system in 6 weeks—enabling engineering to ship for 18 months with 0 additional design hires needed.
- Architected end-to-end product and operational workflows for institutional fiat wires (Fedwire/ACH), segregated multi-sig Bitcoin UTXO custody, rapid interbank settlement (Avit), and internal compliance, BSA/AML, finance, and treasury ops consoles across 100+ rapid prototyping sprints.

PredictiveUX (Contract — Walmart Global Tech) | **Senior UX/UI Architect** | **OCT 2021 - OCT 2022**

- Led UX and information architecture for Walmart's flagship internal enterprise search platform, establishing new search paradigms, GraphQL/graph-database query patterns, and behavioral workflows for 3,000,000+ direct associates.
- Architected clinical decision support and diagnostic workflow systems (RecoverX) translating complex medical ontologies into high-velocity clinician interfaces.

Optum & UnitedHealth Group | **Senior Product Experience Designer** | **JAN 2020 - OCT 2021**

- Operated within an elite internal McKinsey-style executive consulting and R&D group, facilitating complex service design workshops for C-suite and VP leadership across Fortune 5 healthcare operations.
- Served as Principal Designer for flagship native iOS and Android applications spanning 35M+ patients and 400K+ employees, maintaining direct ownership of 80% of Android screens and 50% of iOS screens while stabilizing a 75M-user ecosystem after an unexpected team transition.
- Championed UnitedHealth Group's adoption of native Apple Health guidelines and HealthKit integrations over brittle custom builds, generating an estimated $100M+ in personal revenue impact.

BLDG-25 | **Design Director** | **JAN 2018 - DEC 2019**

- Directed a team of 3 designers and managed key enterprise accounts, including leading the complete modernization of Martin Marietta's nationwide internal inventory and logistics tooling.
- Embedded for 2 years at Fidelity Investments' Raleigh R&D center as primary client liaison, facilitating executive workshops and architecting their first cloud-native application launch.
- Designed mission-critical roles, permissions, and threat-intelligence frameworks for IronNet Cybersecurity and General Keith Alexander (former NSA Director), impacting nation-state level network defense globally.

UltraLinq Healthcare Solutions | **Design Director** | **JAN 2016 - DEC 2017**

- Founded and managed the internal product design team, building zero-to-one design systems across cloud web applications, diagnostic medical imaging tele-health platforms, and wearable cardiac monitors.
- Shipped multiple HIPAA-certified hardware and software medical products, leading contextual clinical research in hospital environments and formal FDA/healthcare compliance testing.

Foodler.com (Acquired by Grubhub) | **Director of Business Operations** | **JAN 2010 - DEC 2015**

- Promoted from Customer Service Representative to Director of Business Operations within 18 months, overseeing two-thirds of company operations including billing, customer support, restaurant ops, and tier 1-2 technical support across $500M+ in cumulative order volume.
- Engineered a proprietary deterministic fraud detection and risk scoring system that reduced credit card chargebacks by 83% (from $6M to under $1M annually) while coordinating directly with local police departments and the FBI on cyber-fraud investigations.
- Scaled operational unit economics and automated merchant/delivery workflows, directly contributing to a 10x revenue expansion and successful acquisition by Grubhub.

EDUCATION

Master of Business Administration (MBA) — Business Strategy & Accounting | University of North Carolina at Chapel Hill (Kenan-Flagler Business School)

Master of Fine Arts (MFA) — Storytelling & Creative Writing | Emerson College

Bachelor of Arts (BA) — Writing & Literature in Honors Studies, Summa Cum Laude | Texas Tech University